

(Time: 2 hours)

[Total Marks: 60]

- N. B.: (1) **All** questions are **compulsory**.
 (2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
 (3) Answers to the **same question** must be **written together**.
 (4) Numbers to the **right** indicate **marks**.
 (5) Draw **neat labeled diagrams** wherever **necessary**.
 (6) Use of **Non-programmable** calculator is **allowed**.

1. **Attempt any two of the following:** 12
 - a. “Any sufficiently advance technology in indistinguishable from magic.” Explain.
 - b. What is manufactured normalcy field? Explain.
 - c. “Data available through IOT device belongs to public or company which implement the IOT device”. Discuss.
 - d. How are dynamic IP addresses assigned by DHCP? Explain.

 2. **Attempt any two of the following:** 12
 - a. Discuss the trade- off between cost and ease of prototyping.
 - b. How can one tap into the community for promoting IoT devices? Explain.
 - c. Explain the process of development on Raspberry Pi .
 - d. What are microcontrollers? Explain.

 3. **Attempt any two of the following:** 12
 - a. How are the following constructed using laser cutter?
 i) Lattice hinges ii) Integrated elastic clips iii) Bolted tenon joints
 - b. What is milling? What is its use? Explain.
 - c. Explain polling. What is its used?
 - d. Discuss the Extensible messaging and presence protocol in detail.

 4. **Attempt any two of the following:** 12
 - a. How can we make optimum use of RAM while writing code for embedded devices?
 - b. What are libraries? Explain with examples
 - c. Define business model. Explain different factors in the definition. What are the different questions that are need to be answered while designing the business model?
 - d. What is venture capital? How can one exit?

 5. **Attempt any two of the following:** 12
 - a. What are the different software choices to design PCB? Explain each in brief.
 - b. What is the importance of certification? Why is it required?
 - c. “The internet destroys the state”. Discuss
 - d. Discuss cautious optimism.
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